

Phase 7 — Regime HMM



Background: Markets switch between hidden regimes such as bull, bear, calm, and crisis.

Definition: Introduce a latent state $R_n \in \{1, \dots, K\}$ evolving as a Markov chain.

Intuition: Use posterior regime probabilities to widen the funnel in unstable regimes and tighten it in stable regimes.

Formulation:

$$r_n | R_n = j \sim \mathcal{N}(\mu_j, \sigma_j^2)$$

$$P(R_n = j | r_{1:n})$$

$$k_n = \sum_j z_{n,j} \mu_j, \quad \delta_n = \sum_j z_{n,j} \sigma_j^2$$

Solution: Estimate $\{\mu_j, \sigma_j^2, A_{ij}\}$ by Baum-Welch / EM on the training data; use forward filtering online on the test fold.

Implementation: Fit HMM on each walk-forward training window, infer regime probabilities live, and set regime-weighted drift / volatility / thresholds into the funnel.

★ *Key idea: HMM is a discrete-state Bayesian filter that complements the Kalman filter.*

Adaptive Funnel Research Roadmap

From rule-based trading exits to regime-aware stochastic control

Current Baseline: Adaptive Funnel V1



Background: A trading-exit framework that avoids voluntary realized loss on completed algorithmic exits.

Core objects: X_t = log-price, C_t = predictive center, V_t = cumulative predictive variance

Normalized deviation: $Z_t = \frac{X_t - C_t}{\sqrt{V_t}}$

Profit buffer: $\Pi_t = X_t - X_b - \ell_c, \quad \ell_c = \log\left(\frac{1 + c_{tx}}{1 - c_{cell}}\right)$

Sell rule: Sell when $Z_t > k_t$, momentum weakens, and $\Pi_t > 0$

Intuition: Take profit only when price is sufficiently above model expectation, continuation weakens, and post-cost profit is positive.

🏆 *Goal: keep the no-voluntary-realized-loss guarantee while making the strategy smarter and more optimal.*

Phase 8 — Randomized Stopping



Background: Hard thresholds can be brittle and overly sensitive near the boundary.

Definition: Replace the deterministic sell trigger with a probability of selling.

Intuition: Sell softly near the boundary; recover the original rule as a limiting case.

Formulation:

$$p_t = \sigma(a(Z_t - k_t) - bG_t + c\Pi_t)$$

$$D_t \sim \text{Bernoulli}(p_t)$$
 Sell when $D_t = 1$, while enforcing $\Pi_t > 0$. As $a \rightarrow \infty$, logistic policy $\rightarrow$ deterministic rule.

Solution: Choose policy parameters by maximizing utility or balancing reward, holding time, and risk.

Implementation: Run Monte Carlo backtests, compare deterministic vs probabilistic exits, and always enforce $\Pi_t > 0$ to preserve the guarantee.

★ *Extra lens: hazard rate and survival analysis for holding-time distribution.*

Phase 9 — Generalized Momentum



Background: A single momentum statistic is often too crude.

Definition: Build a continuation score from multiple features instead of one raw difference.

Intuition: The funnel stays where price is, generalized momentum says whether the move still has life.

Formulation:

$$\phi_t = [\text{short momentum, medium momentum, } \hat{\mu}_t, \hat{\sigma}_t - \hat{\sigma}_t, Z_{t-6}, \text{drawdown, regime probs}]$$

$$G_t = g(\phi_t)$$
 Sell if $G_t < \vartheta$ and $\Pi_t > 0$
 Time-decaying threshold: $\vartheta(\tau) = \vartheta_0 e^{-\tau\tau_0}$

Solution: Use a weighted slope estimator, linear score, or logistic continuation model.

Implementation: Engineer features, estimate θ , compare finite-difference momentum vs slope-based momentum, and validate out-of-sample.

★ *Elegant symmetry: funnel width grows like $\sqrt{V_t}$, while momentum tolerance relaxes like $1/\sqrt{t}$.*

Phase 10 — HJB / Free-Boundary Theory



Background: Provide a stochastic-control foundation for the adaptive funnel.

Definition: Formulate selling as a constrained optimal stopping problem with transaction costs and holding-time penalty.

Intuition: The adaptive funnel acts as a tractable approximation to an optimal free boundary.

Formulation:

$$V(t, x) = \sup E[e^{-\rho t} G(X_t) - \lambda \int_t^T e^{-\rho s} ds]$$
 Variational inequality: $\min\{-\mathcal{L}V - \lambda, V - G\} = 0$

$$\mathcal{L} = \frac{\partial}{\partial t} + \mu \frac{\partial}{\partial x} + \frac{1}{2} \sigma^2 \frac{\partial^2}{\partial x^2}$$
 Free boundary $x^*(t)$: $V(t, x^*) = G(x^*)$, $\mathcal{L}V(t, x^*) = G'(x^*)$

Solution: Solve the finite-horizon variational inequality numerically; approximate solution via omni-calms.

Implementation: Use finite differences, policy iteration, or penalty methods; compare the computed free boundary with the adaptive funnel boundary; benchmark to lower-bound for exit and re-entry.

★ *Capstone insight: the free-boundary can be interpreted as an approximate optimal boundary.*



Unified Vision

Adaptive Funnel V2/V3 = state estimation + regime awareness + continuation scoring + constrained stopping + optimal-control interpretation

This roadmap showcases mathematics, statistics, stochastic processes, and implementation maturity.


